ID:	Name:

Brac University
Semester: Summer 2025
Course Code: CSE251
Electronic Circuits And Devices
Section: 6
Faculty: PDS

Set
B

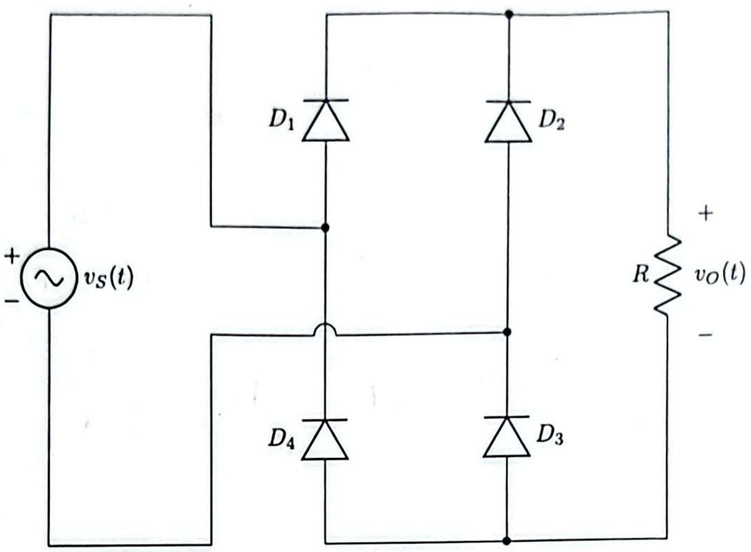
Assessment: Quiz 3
Duration: 30 minutes
Date: August 24, 2025
Full Marks: 20

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **3 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

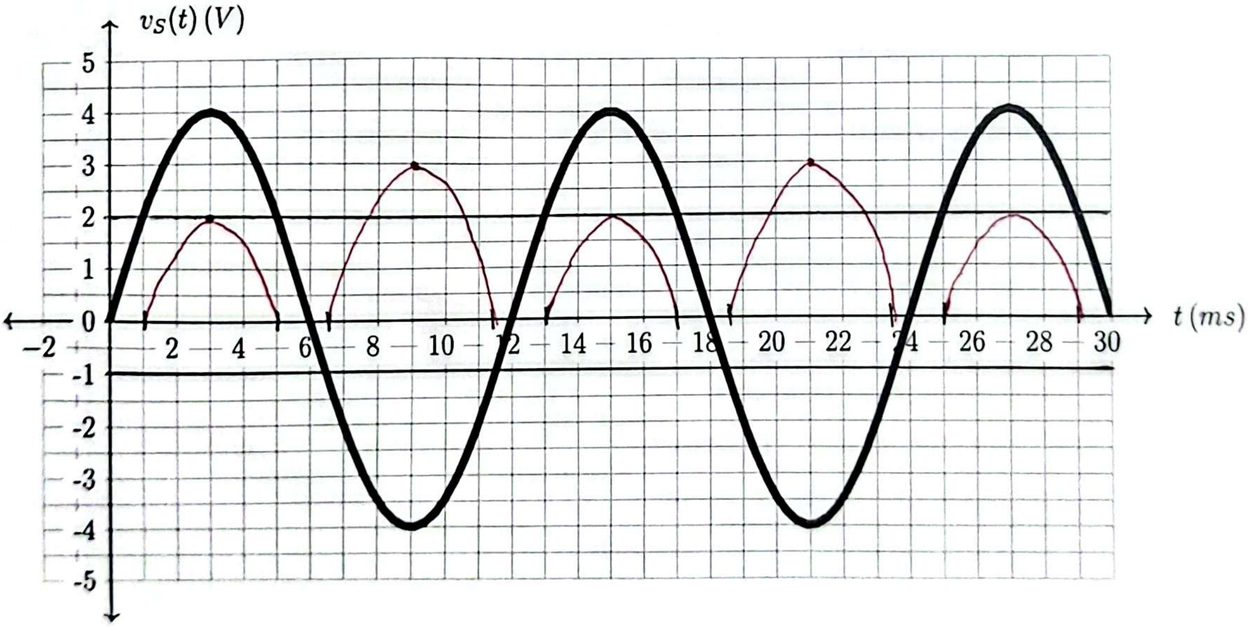
■ Question 1 of 3 *[CO2] [4 marks]*

For the rectifier circuit shown, the diodes D_1, D_2, D_3, and D_4 have cut-in voltages of 1.2 V, 0.5 V, 0.8 V, and 0.5 V, respectively. The input voltage v_S is a sinusoidal waveform as plotted below. Sketch the output voltage v_O on the same grid as v_S.

(−) half cycles
D_2 & D_4 ON
voltage drop = 0·5 + 0·5 = 1V



(+) half cycles
D_1 & D_3 ON
Voltage drop = 1·2 + 0·8 = 2V



■ Question 2 of 3 [CO2] [6 marks]

A 12 V amplitude, 100 Hz sinusoidal AC voltage is rectified and filtered using a FW rectifier with 4 μF filter capacitor. The diodes used in the rectifier are identical and have a cut-in voltage of 1.2 V. If the average output voltage is measured to be 9 V, determine the value of the load resistance.

$V_m = 12V$

$f = 100 Hz$

$C = 4 \times 10^{-6} F$

$V_{D_0} = 1.2V$

$V_{avg} = V_{DC} = 9V$

$V_p = V_m - 2V_{D_0} = 9.6V$

$V_{avg} = V_p - \frac{V_r}{2}$

$\Rightarrow V_r = 2(V_p - V_{avg})$

$= 2(9.6 - 9) = 1.2V$

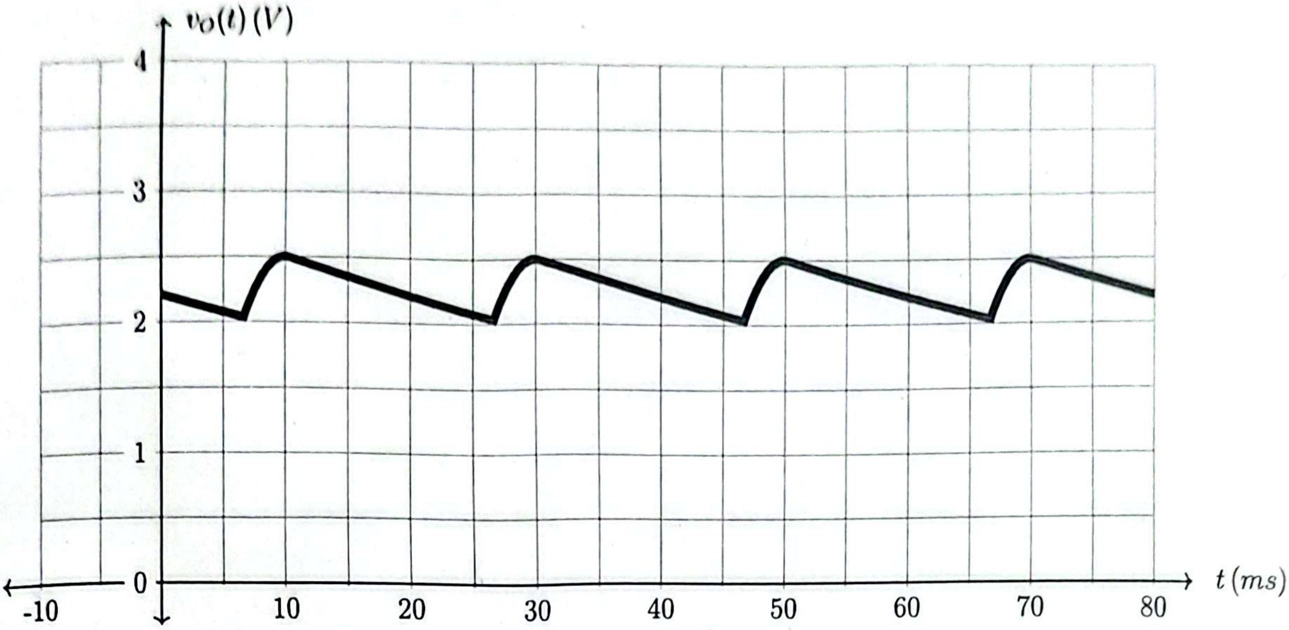
$V_r = \frac{V_p}{2fRC}$

$\Rightarrow 1.2 = \frac{9.6}{2 \times 100 \times R \times 4 \times 10^{-6}}$

$\Rightarrow \boxed{R = 10 k\Omega}$

■ Question 3 of 3 *[CO3] [10 marks]*

The following is the rectified output of an AC sinusoidal voltage at 25 Hz frequency, applied to a load of 20 kΩ. The diode(s) used in the rectifier have a cut-in voltage of $V_{D0} = 0.6$ V.



(a) *[2 marks]* What type of rectifier was used?

(b) *[4 marks]* What is the filter capacitance value used in the rectifier design?

(c) *[2 marks]* Write an expression for the input voltage as a function of time t.

(d) *[2 marks]* Draw the circuit diagram of the rectifier.

(a) Time period of output = 30 − 10 = 20ms

Frequency of output = $\frac{1}{20ms}$ = 50HZ

As output frequency = 2 × Input frequency,

It was $\boxed{\text{FW rectifier}}$

(b) From the plot,

$V_P = 2.5V$, $V_R = 0.5V$

$f = 25\,HZ$

$R = 20\,K\Omega$

$V_{D0} = 0.6V$

$$V_R = \frac{V_P}{2fRC}$$

$$\Rightarrow 0.5 = \frac{2.5}{2 \times 25 \times 20 \times 10^3 \times C} \Rightarrow \boxed{C = 5\mu F}$$

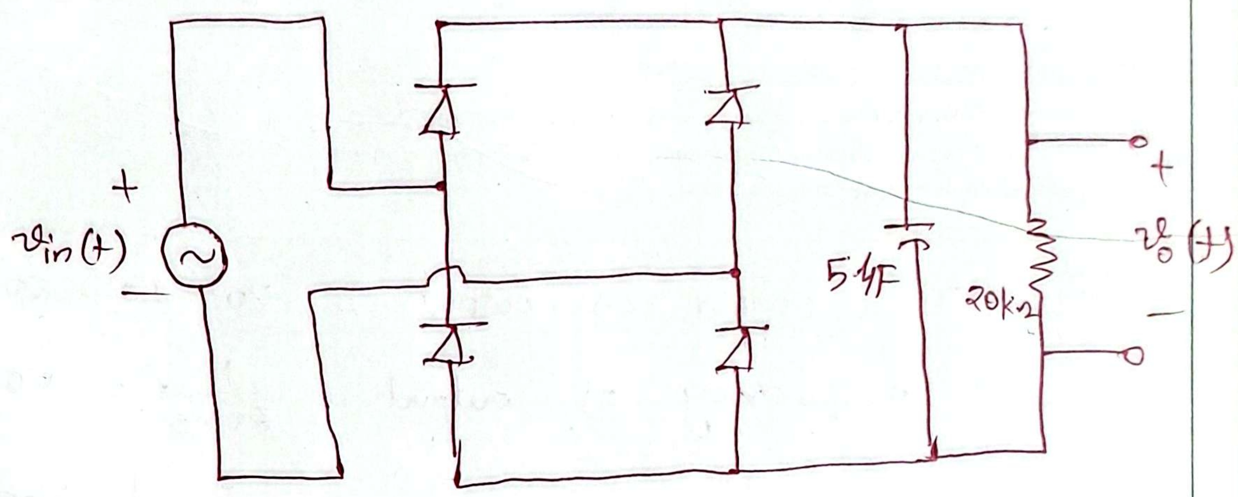
(c) $v_{in}(t) = V_m \sin(2\pi f t)$ (V)

$V_m = V_p + 2V_{D_0} = 2.5 + 2 \times 0.6$

$= 3.7 V$

$$\boxed{v_{in}(t) = 3.7 \sin(2\pi 25 t) \text{ (V)}}$$

(d)



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A

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Assessment: Quiz 7
Duration: 30 minutes
Date: August 24, 2025
Full Marks: 20

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ **All 3 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 3 *[CO2] [4 marks]*

For the rectifier circuit shown, the diodes D_1, D_2, D_3, and D_4 have cut-in voltages of 0.5 V, 1.2 V, 0.5 V, and 0.8 V, respectively. The input voltage v_S is a sinusoidal waveform as plotted below. Sketch the output voltage v_O on the same grid as v_S.

(−) half cycles

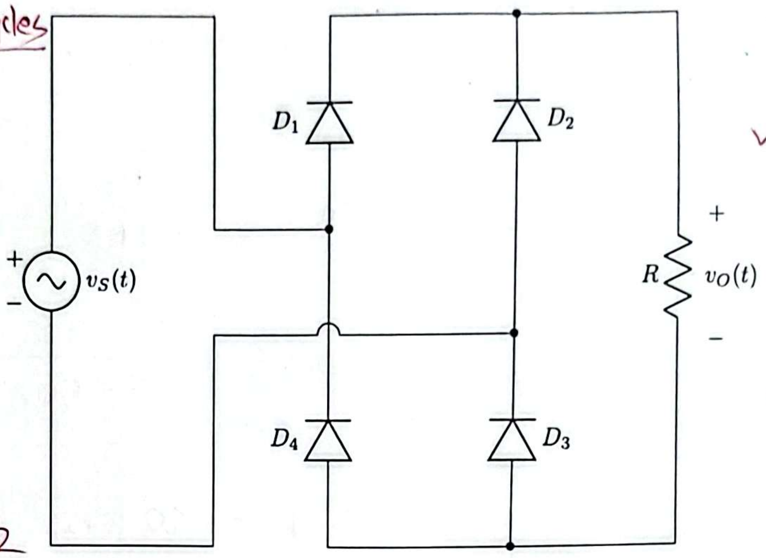
D_2 & D_4 ON

Voltage drop = $1.2+0.8$

$= 2V$

$V_p = V_m - 2$

$= 2V$



(+) half cycles

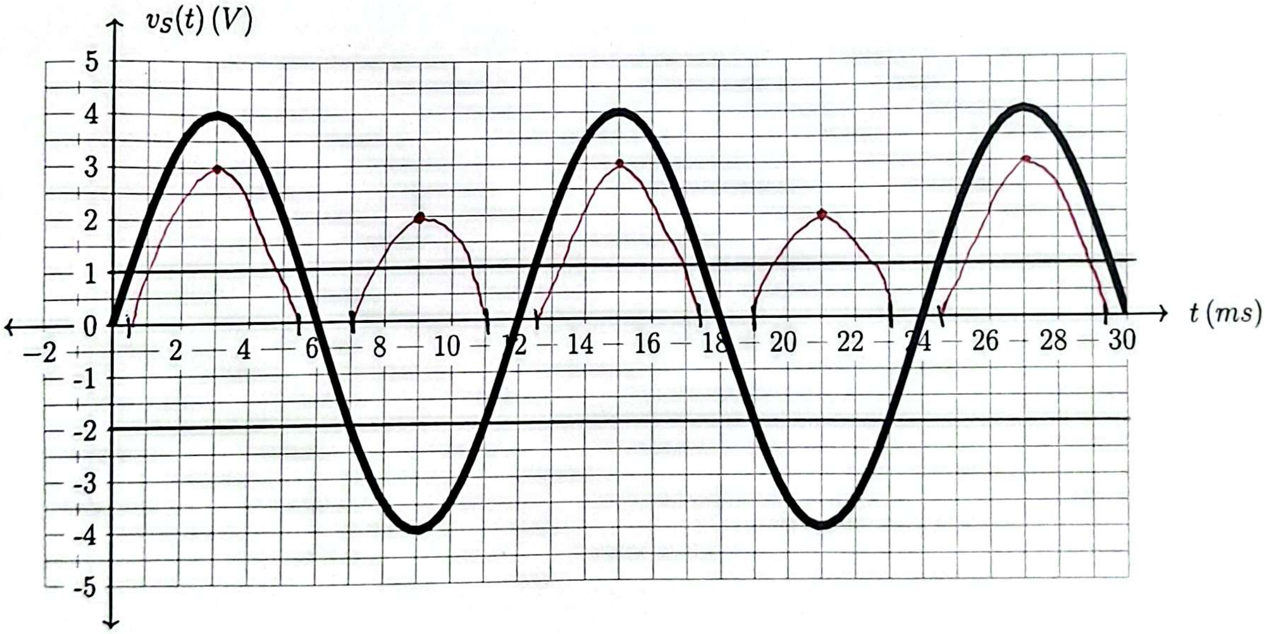
D_1 & D_3 ON

Voltage drop $= 0.5+0.5$

$= 1V$

$V_p = V_m - 1$

$= 3V$



■ Question 2 of 3 [CO2] [6 marks]

A 6 V amplitude, 50 Hz sinusoidal AC voltage is rectified and filtered using a FW rectifier with 8 μF filter capacitor. The diodes used in the rectifier are identical and have a cut-in voltage of 0.6 V. If the average output voltage is measured to be 4.5 V, determine the value of the load resistance.

$V_m = 6V$

$f = 50Hz$

$C = 8 \times 10^{-6} F$

$V_{D0} = 0.6V$

$V_{avg} = V_{DC} = 4.5V$

$V_p = V_m - 2V_{D0} = 4.8V$

$V_{avg} = V_p - \frac{V_r}{2}$

$\Rightarrow V_r = 2(V_p - V_{avg})$

$= 2(4.8 - 4.5)$

$= 0.6V$

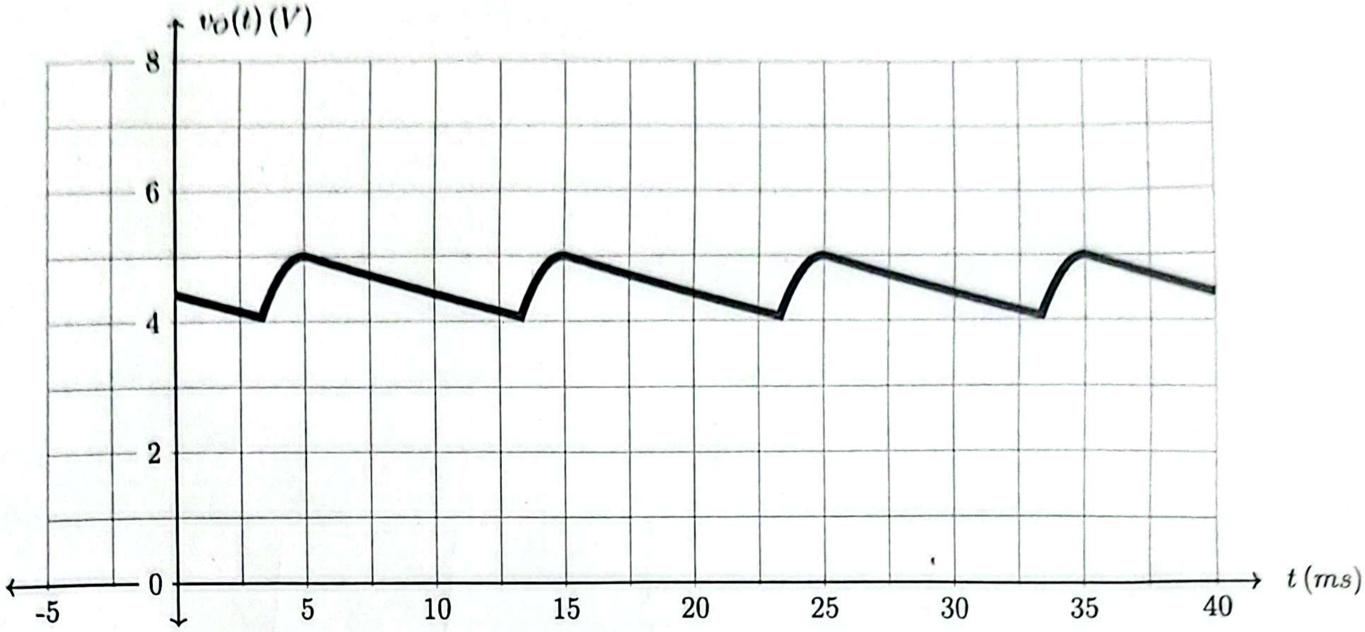
$V_r = \frac{V_p}{2fRC}$

$\Rightarrow 0.6 = \frac{4.8}{2 \times 50 \times R \times 8 \times 10^{-6}}$

$\Rightarrow R = 10 K\Omega$

■ Question 3 of 3 [CO3] [10 marks]

The following is the rectified output of an AC sinusoidal voltage at 50 Hz frequency, applied to a load of 10 kΩ. The diode(s) used in the rectifier have a cut-in voltage of $V_{D0} = 0.6$ V.



(a) *[2 marks]* What type of rectifier was used?
(b) *[4 marks]* What is the filter capacitance value used in the rectifier design?
(c) *[2 marks]* Write an expression for the input voltage as a function of time t.
(d) *[2 marks]* Draw the circuit diagram of the rectifier.

(a) Time period of output = 15 − 5 = 10 ms

Frequency of output $= \frac{1}{10\text{ms}} = 100$ Hz

As output frequency = 2 × Input frequency

It was $\boxed{\text{FW rectifier}}$

(b) From the plot,

$f = 50$ Hz
$R = 10$ kΩ
$V_{D0} = 0.6$ V

$V_p = 5$ V, $V_r = 1$ V

$$V_r = \frac{V_p}{2fRC}$$

$$\Rightarrow 1 = \frac{5}{2 \times 50 \times 10 \times 10^3 \times C} \Rightarrow \boxed{C = 5\,\mu F}$$

(c) $v_{in}(t) = V_m \sin(2\pi f t)$

$$V_m = V_p + 2V_{D0} = 5 + 2\times 0.6$$
$$= 6.2V$$

$$v_{in}(t) = 6.2 \sin(2\pi 50 t) \text{ (V)}$$

(d)